

NOTTINGHAM WATNALL, United Kingdom

WMO: 033540

Lat: 53.0056N

Lon: 1.2511W

Elev: 117

StdP: 99.93

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.9	-1.8	-6.1	2.3	-0.7	-4.7	2.6	-0.1	11.1	7.9	10.2	8.0	2.9	40	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	26.3	18.4	24.1	17.3	22.2	16.5	19.3	24.1	18.2	22.5	17.3	21.1	3.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	12.8	21.0	16.6	12.0	19.8	15.7	11.3	18.8	55.3	24.3	51.8	22.6	49.0	21.2	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.2	8.2	7.3	-5.2	29.8	1.9	2.1	-6.6	31.3	-7.7	32.5	-8.8	33.6	-10.2	35.1
			-5.7	21.6	1.9	1.4	-7.1	22.6	-8.2	23.4	-9.3	24.2	-10.6	25.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	4.3	4.7	6.3	8.7	11.6	14.5	16.8	16.5	13.9	10.7	7.0	4.7
	DBStd	5.32	2.98	3.12	2.96	2.96	2.97	2.68	2.66	2.40	2.34	2.74	2.95	3.40
	HDD10.0	809	178	149	119	59	16	1	0	0	1	23	97	166
	HDD18.3	3073	436	382	373	290	209	118	65	68	133	236	341	423
	CDD10.0	813	1	1	4	19	66	137	209	202	119	46	6	2
	CDD18.3	34	0	0	0	0	1	4	16	12	2	1	0	0
	CDH23.3	261	0	0	0	1	6	41	127	73	12	1	0	0
	CDH26.7	43	0	0	0	0	0	7	22	14	1	0	0	0
Wind	WSAvg	3.6	4.1	4.1	4.0	3.6	3.5	3.2	3.0	3.0	3.1	3.6	3.6	3.9
Precipitation	PrecAvg	715	65	50	52	51	53	57	56	61	61	61	65	71
	PrecMax	937	127	198	114	115	135	164	147	128	143	144	132	180
	PrecMin	521	16	10	6	7	11	5	11	6	9	10	25	19
	PrecStd	106	26	36	27	27	31	37	33	29	37	34	26	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.4	13.6	16.1	21.2	24.1	27.5	29.4	28.8	25.0	19.7	15.1	13.4
		MCWB	10.9	10.9	11.1	13.7	16.6	18.8	19.5	19.2	17.6	16.1	13.5	11.6
	2%	DB	11.1	11.6	13.8	17.6	21.6	24.0	26.5	25.5	21.7	17.3	13.6	12.2
		MCWB	9.5	9.6	10.2	12.2	15.3	16.9	18.4	18.5	16.7	14.5	12.2	10.9
	5%	DB	9.9	10.5	12.3	15.5	19.2	21.7	24.4	22.9	19.7	15.9	12.5	11.0
		MCWB	8.6	8.7	9.4	11.0	13.9	15.8	17.5	17.1	15.4	13.7	11.1	9.7
	10%	DB	8.9	9.3	11.0	13.8	17.1	19.9	22.2	21.2	18.2	14.8	11.4	9.7
		MCWB	7.6	7.7	8.5	10.1	12.7	15.0	16.6	16.3	14.5	13.0	10.1	8.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	11.3	12.0	14.6	17.7	20.1	21.0	20.9	18.8	16.6	14.0	12.0
		MCDB	12.2	12.8	14.5	19.8	22.4	26.0	27.4	25.5	22.8	19.1	14.8	12.9
	2%	WB	9.8	9.9	10.9	12.9	15.9	17.8	19.2	19.2	17.4	15.3	12.6	11.1
		MCDB	10.9	11.4	12.9	16.5	20.7	22.4	24.6	23.1	20.6	16.7	13.4	12.1
	5%	WB	8.8	8.9	9.8	11.7	14.6	16.6	18.1	18.0	16.2	14.2	11.4	10.0
		MCDB	9.7	10.3	11.9	14.9	18.3	20.5	22.7	22.1	18.7	15.5	12.2	10.9
	10%	WB	7.7	7.9	8.9	10.5	13.3	15.6	17.2	17.0	15.3	13.2	10.3	8.7
		MCDB	8.8	9.1	10.7	13.3	16.3	19.0	21.2	20.3	17.4	14.5	11.2	9.6
Mean Daily Temperature Range	5% DB	MDBR	4.6	5.3	6.6	7.9	8.4	8.3	8.7	8.3	7.6	6.2	5.1	4.5
		MCDBR	5.8	6.4	8.4	10.4	11.6	11.7	12.3	11.1	9.8	7.0	5.7	5.7
		MCWBR	5.4	4.9	5.5	6.0	6.5	6.1	5.5	5.1	5.3	4.7	4.9	5.2
	5% WB	MCDBR	5.7	6.0	7.6	9.8	10.7	10.5	10.9	9.6	8.7	6.6	5.6	5.5
		MCWBR	5.5	5.0	5.5	6.1	6.4	6.3	5.6	5.0	5.6	5.1	5.1	5.2
Clear-Sky Solar Irradiance	taub	0.321	0.336	0.368	0.395	0.398	0.399	0.399	0.386	0.367	0.348	0.323	0.309	
	taud	2.349	2.386	2.295	2.232	2.263	2.284	2.288	2.349	2.406	2.437	2.405	2.328	
	Ebn at Noon	654	768	812	834	851	852	845	836	808	744	651	592	
	Edh at Noon	60	77	103	124	126	125	122	109	92	73	56	51	
All-Sky Solar Radiation	RadAvg	0.71	1.40	2.47	3.80	4.73	4.98	4.82	3.97	2.95	1.70	0.88	0.57	
	RadStd	0.07	0.16	0.30	0.44	0.31	0.48	0.51	0.27	0.27	0.15	0.10	0.06	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (81 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page